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June 13, 2019
Pulte Tucson Land - MS-1021
P.O. Box 3660
Portland, OR 97208

Re: Building Pad-Post-Tension Foundation [102003477]

5X and 5W Mass Grading @ Rancho Vistoso Valley Vista
Lot #: 1-47, 98-127 (Phase 1A) Permit #:

ProTeX Job ID#: 0234Dev-3579-8944

To Whom It May Concern:

In accordance with your request ProTeX has completed observation, moisture and density testing as a part of the construction of the building pads associated with the aforementioned development. Surface clearing, over-excavation, preparation of subgrade, and compaction of subgrade/fill soils are in accordance with the specifications included in the Soil Investigation conducted by ProTeX the PT Xperts, LLC., (Job No. 7466), dated February 20, 2018.

Field observations were made to ensure that the site was cleared of all vegetation and structures, site was over-excavated to a depth of 1 foot and exposed subgrade soils were scarified. Testing was performed to confirm moisture conditioning and compaction efforts are in compliance with the recommendations of the aforementioned geotechnical report. Field Density tests are enclosed. Based on the observations, laboratory testing, and field density testing, a bearing value of 1250psf at finished grade bearing on 1 foot of engineered fill is assigned for post-tensioned foundation systems. Building Pads are certified for one year from the date of this letter.

Areas in the building pad that are disturbed as a result of construction activities should be re-compacted to specified density. Inspection of foundation bearing soils prior to placement of concrete is recommended. Proper drainage should be maintained at all times away from structures. Following the construction of buildings and prior to occupancy, it is recommended that drainage away from the structure should be inspected under the supervision of a registered professional engineer in the civil or geotechnical field.

Submitted by:
ProTeX - the PT Xperts, L.L.C.



Expires 3/31/2020

Jane Sample , P.E.

Field Density Summary Report

Client: Pulte Tucson Land
Job Name: 5X and 5W Mass Grading
Job ID #: 8944

Project Name: Rancho Vistoso Valley Vista
Cross Streets: La Canada Drive and Moore Road
Lot: Earthwork

Test Type: Pad (5-10 feet) (Phase 1A)

Date	Street	Lot / Sta.	Depth of Test	Direct Trans. Used	Field Dry Density (lb/ft3)	Field Moisture (%)	Max Dry Density (corr) (lt/ft3)	Opt. Moisture (corr) (%)	% Comp.	Density Spec. (%)	Moisture Spec. (%)	Proctor Lab #	Pass
04/17/2019	Auterbrunnen Lane	Lot 39	7' BFG	10"	101.9	9.6	104.1	8.2	98	98 min.	-2 to +2	191996	Pass
04/17/2019	Auterbrunnen Lane	Lot 43	5' BFG	10"	100.4	10.4	102.2	8.6	98	98 min.	-2 to +2	191996	Pass
04/17/2019	Auterbrunnen Lane	Lot 46	6' BFG	10"	107.1	9.3	108.1	7.3	99	98 min.	-2 to +2	191996	Pass
04/18/2019	Auterbrunnen Lane	Lot 37	6' BFG	10"	115.5	8.6	118.4	9.0	98	98 min.	-2 to +2	191999	Pass
04/18/2019	Auterbrunnen Lane	Lot 34	6' BFG	10"	113.2	7.5	114.8	6.0	99	98 min.	-2 to +2	191996	Pass
04/19/2019	Romsdalen Road	Lot 20	8' BFG	10"	123.0	9.8	126.0	9.3	98	98 min.	-2 to +2	191995	Pass
04/19/2019	Romsdalen Road	Lot 25	6' BFG	10"	119.9	8.0	121.2	10.0	99	98 min.	-2 to +2	191957	Pass
04/19/2019	Romsdalen Road	Lot 23	7' BFG	10"	120.4	9.3	121.2	10.0	99	98 min.	-2 to +2	191957	Pass
04/19/2019	Romsdalen Road	Lot 24	8' BFG	10"	120.0	8.9	121.2	10.0	99	98 min.	-2 to +2	191957	Pass
04/19/2019	Romsdalen Road	Lot 21	7' BFG	10"	123.7	8.8	126.0	9.3	98	98 min.	-2 to +2	191995	Pass
04/19/2019	Romsdalen Road	Lot 22	6' BFG	10"	117.1	8.3	118.4	9.0	99	98 min.	-2 to +2	191999	Pass
04/26/2019	Romsdalen Road	Lot 26	5' BFG	10"	125.1	8.0	126.0	9.3	99	98 min.	-2 to +2	191995	Pass
04/26/2019	Romsdalen Road	Lot 20	5' BFG	10"	118.8	8.0	121.2	10.0	98	98 min.	-2 to +2	191957	Pass
04/26/2019	Romsdalen Road	Lot 24	5' BFG	10"	120.8	8.8	121.2	10.0	100	98 min.	-2 to +2	191957	Pass
04/29/2019	Kalalau Drive	Lot 19	7' BFG	10"	116.3	7.6	118.4	9.0	98	98 min.	-2 to +2	191999	Pass
04/29/2019	Kalalau Drive	Lot 19	8' BFG	10"	118.5	7.2	118.4	9.0	100	98 min.	-2 to +2	191999	Pass
04/29/2019	Kalalau Drive	Lot 18	6' BFG	10"	119.7	8.0	118.4	9.0	101	98 min.	-2 to +2	191999	Pass
04/29/2019	Kalalau Drive	Lot 17	5' BFG	10"	117.8	9.9	118.4	9.0	100	98 min.	-2 to +2	191999	Pass

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Soils Summary

Client:	Pulte Tucson Land	ProTeX Job No:	8944
Project Name:	Rancho Vistoso	ProTeX Lab No:	191957 - Phoenix
Job Name:	5X and 5W Mass Grading	Date Received:	3/27/2019
Material:	Native (On-Site)	Sampled By:	John Doe
Material Supplier:	-	Date Sampled:	3/25/2019
Sample Location:	Lot 10 (-)	Submitted By:	John Doe

Plasticity Index

Liquid Limit	**NA**
Plastic Limit	**NA**
Plasticity Index	**NA**

Expansion Index, (EI)	Potential Expansion
0 - 20	Very Low
21 - 51	Low
52 - 90	Medium
91 - 130	High
> 130	Very High

Expansion Index

EI = **NA**

Percent Swell of Soil

% Swell	**NV**

Notes:

pH and Resistivity

pH Reading:	**NA**
Resistivity (ohms-cm)	**NA**

Class:

Symbol:

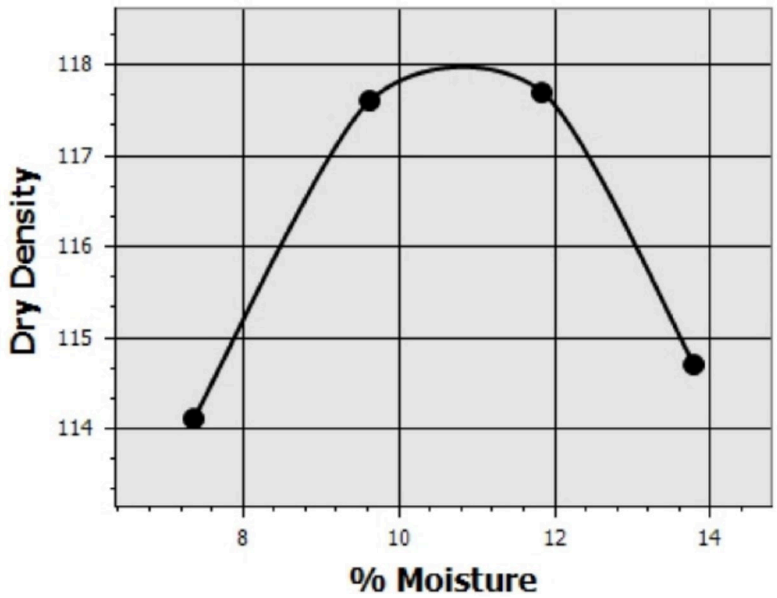
ASTM D698-A

Moisture Density (Proctor)

Max. Dry Density	**118.0**
Opt. Moisture %	**10.8**
Corr. Max. Dry Density	**121.2**
Corr. Opt. Moisture %	**10.1**
% Rock	**8**

* = out of specification

Moisture Vs. Density



Remarks: Grus with sand

Reviewed By: R Roe

Richard Roe

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Soils Summary

Client:	Pulte Tucson Land	ProTeX Job No:	8944
Project Name:	Rancho Vistoso	ProTeX Lab No:	191995 - Phoenix
Job Name:	5X and 5W Mass Grading	Date Received:	3/28/2019
Material:	Native (On-Site)	Sampled By:	John Doe
Material Supplier:		Date Sampled:	3/25/2019
Sample Location:	Lot 53 (#7) (-)	Submitted By:	John Doe

Plasticity Index

Liquid Limit	NA
Plastic Limit	NA
Plasticity Index	NA

Expansion Index, (EI)	Potential Expansion
0 - 20	Very Low
21 - 51	Low
52 - 90	Medium
91 - 130	High
> 130	Very High

Expansion Index

EI = NA

Percent Swell of Soil

% Swell	NV

Notes:

pH and Resistivity

pH Reading:	NA
Resistivity (ohms-cm)	NA

Class:

Symbol:

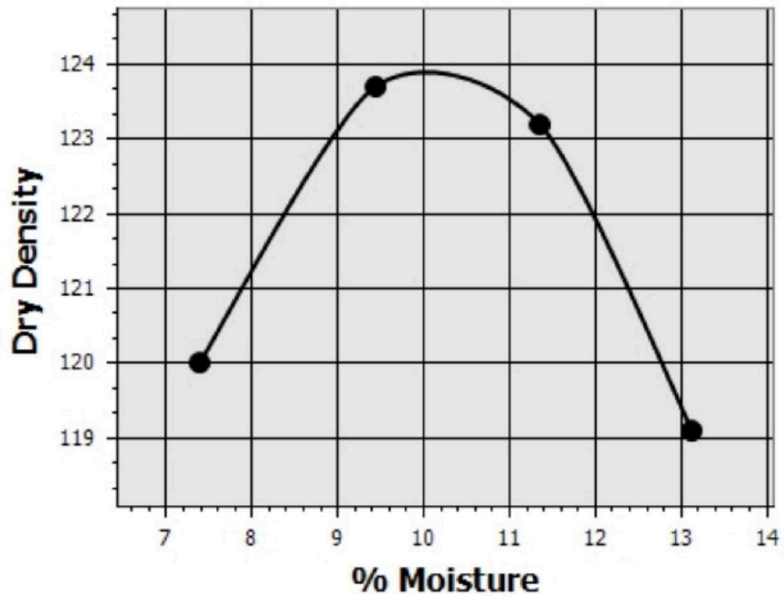
ASTM D698-A

Moisture Density (Proctor)

Max. Dry Density	123.9
Opt. Moisture %	10.0
Corr. Max. Dry Density	126.0
Corr. Opt. Moisture %	9.5
% Rock	7

* = out of specification

Moisture Vs. Density



Remarks:

Reviewed By: R Roe

Richard Roe

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Soils Summary

Client:	Pulte Tucson Land	ProTeX Job No:	8944
Project Name:	Rancho Vistoso	ProTeX Lab No:	191996 - Phoenix
Job Name:	5X and 5W Mass Grading	Date Received:	3/28/2019
Material:	Native (On-Site)	Sampled By:	John Doe
Material Supplier:	-	Date Sampled:	3/25/2019
Sample Location:	Lot 127 (#2) (-)	Submitted By:	John Doe

Plasticity Index

Liquid Limit	NA
Plastic Limit	NA
Plasticity Index	NA

Expansion Index, (EI)	Potential Expansion
0 - 20	Very Low
21 - 51	Low
52 - 90	Medium
91 - 130	High
> 130	Very High

Expansion Index

EI = NA

Percent Swell of Soil

% Swell	NV

Notes:

pH and Resistivity

pH Reading:	NA
Resistivity (ohms-cm)	NA

Class:

Symbol:

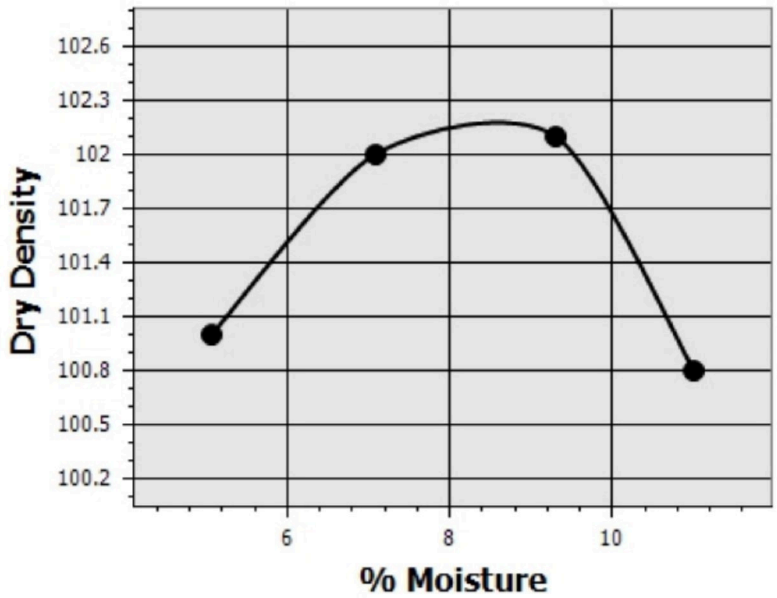
ASTM D698-A

Moisture Density (Proctor)

Max. Dry Density	102.2
Opt. Moisture %	8.6
Corr. Max. Dry Density	104.1
Corr. Opt. Moisture %	8.2
% Rock	5

* = out of specification

Moisture Vs. Density



Remarks:

Reviewed By: R Roe

Richard Roe

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Soils Summary

Client:	Pulte Tucson Land	ProTeX Job No:	8944
Project Name:	Rancho Vistoso	ProTeX Lab No:	191999 - Phoenix
Job Name:	5X and 5W Mass Grading	Date Received:	3/28/2019
Material:	Native (On-Site)	Sampled By:	John Doe
Material Supplier:		Date Sampled:	3/25/2019
Sample Location:	Lot 142 (#9) (-)	Submitted By:	John Doe

Plasticity Index

Liquid Limit	NA
Plastic Limit	NA
Plasticity Index	NA

Expansion Index, (EI)	Potential Expansion
0 - 20	Very Low
21 - 51	Low
52 - 90	Medium
91 - 130	High
> 130	Very High

Expansion Index

EI = NA

Percent Swell of Soil

% Swell	NV

Notes:

pH and Resistivity

pH Reading:	NA
Resistivity (ohms-cm)	NA

Class:

Symbol:

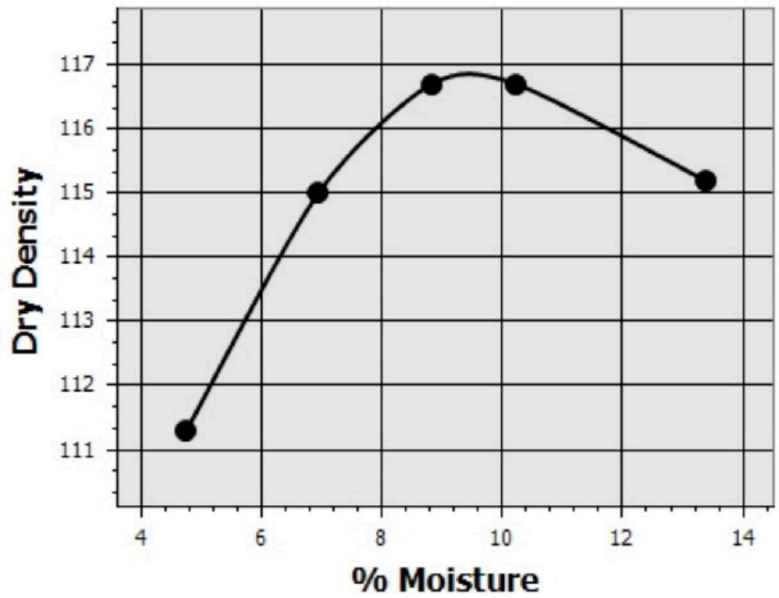
ASTM D698-A

Moisture Density (Proctor)

Max. Dry Density	116.9
Opt. Moisture %	9.5
Corr. Max. Dry Density	118.4
Corr. Opt. Moisture %	9.1
% Rock	5

* = out of specification

Moisture Vs. Density



Remarks:

Reviewed By: R Roe

Richard Roe